Assembling the PCBs

General information

Since I cannot recommend this project for beginners anyway (high voltage, surface mount, lots of solder joints, etc.), I will skip all the basic stuff and explain just a few things that might be of interest.

Controller

Start assembling the input protection section including the LED and fuse. Install a jumper on the SWITCH header. According to the datasheet the circuit should protect against overvoltage up to 60V and against reverse voltage down to -40V.

Connect the board to a variable power supply. Start with 12V, the LED should turn on. If you increase the voltage, the LED should turn off around 13.5V. If you decrease the voltage, the LED should turn off around 10V. These are only approximate values.

Then assemble the +3.3V and the +5V supplies and check that all voltages are correct.

Don't populate the SERIAL header.

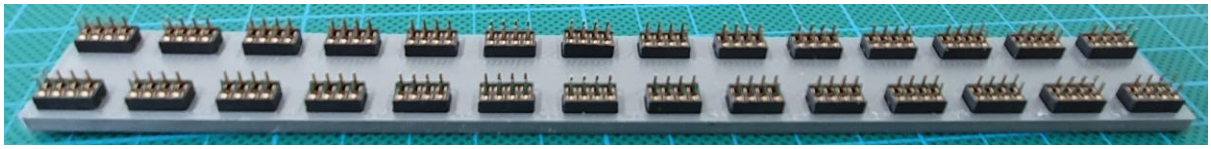
Keyboard

- Solder everything except D1-D35, K1-K35, SW1, H1 and ON
- Flash the bootloader (see Docs/Flashing the bootloader and the firmware.pdf)
- Flash the firmware
- Solder the remaining components

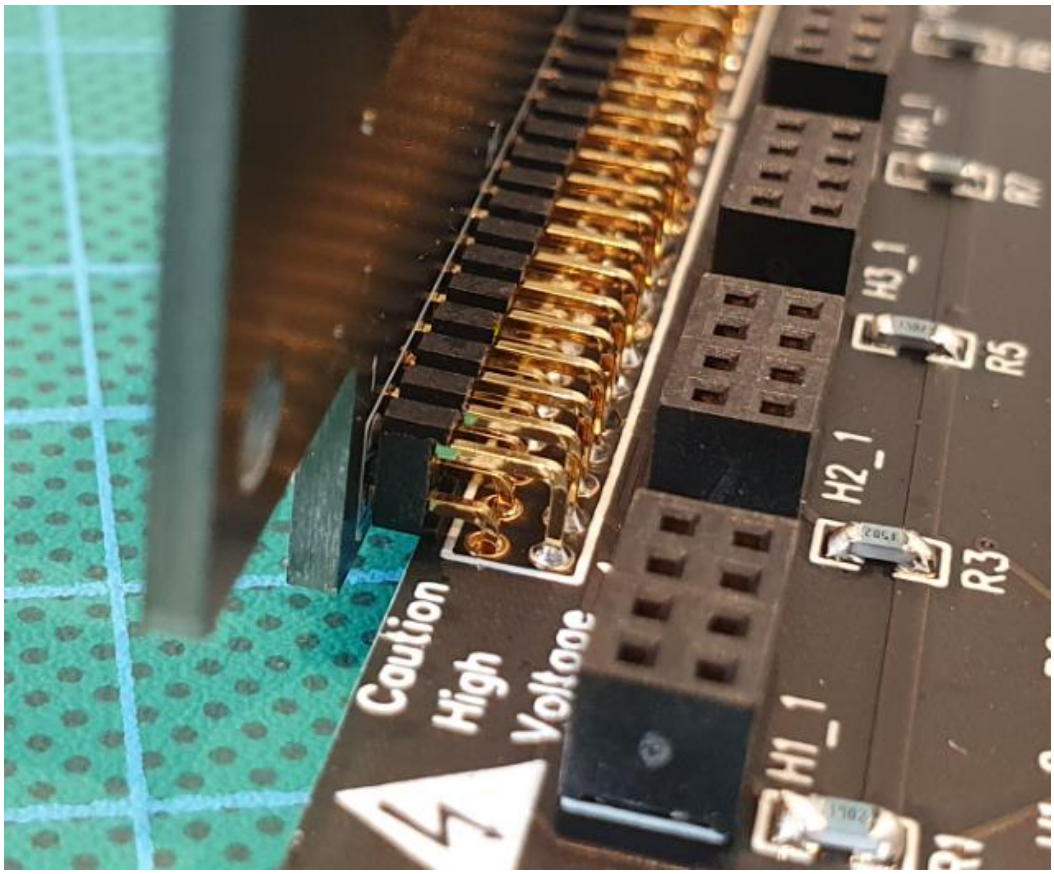
Display and driver boards

IN-16 Version

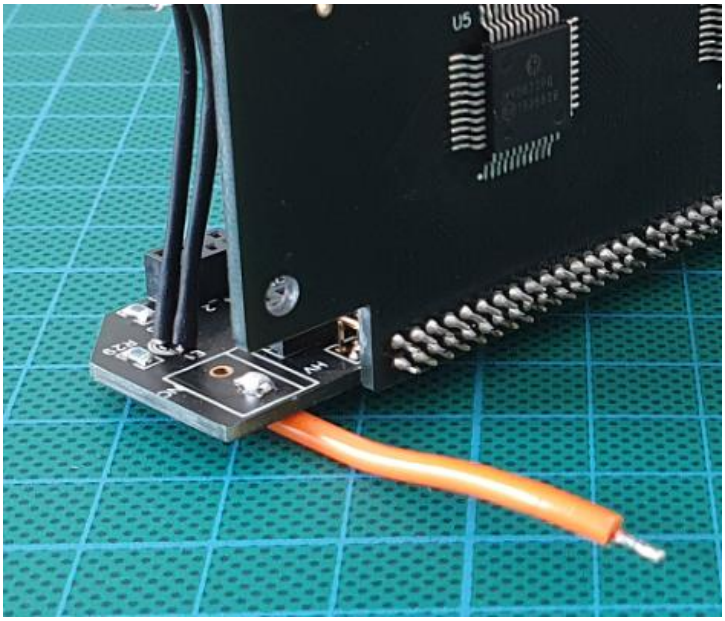
The IN-16 display tool can be used to keep things aligned, assuming you use header sockets with similar dimensions. Don't use too much solder when soldering the socket headers or solder may flow into the sockets.



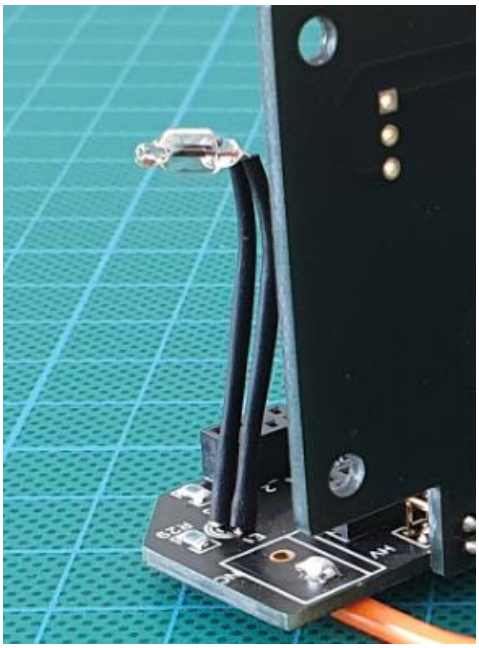
Driver and display boards must be soldered together as shown in the picture using right angle headers.



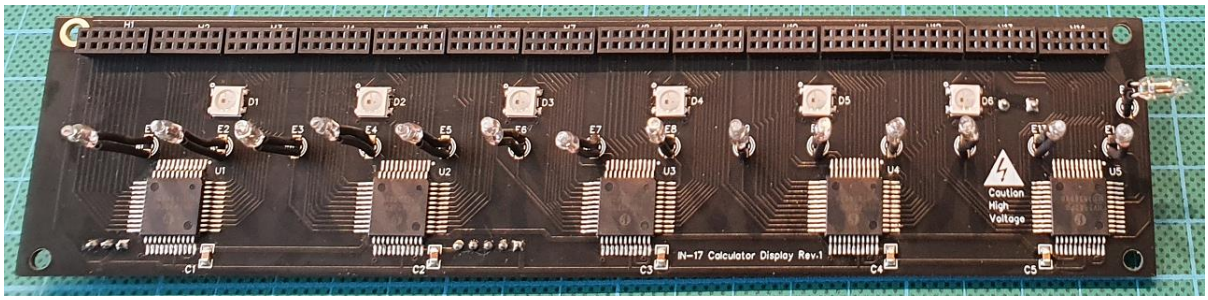
To connect the display board to the HV power supply I have soldered a wire directly to the board, but you can also use a 2-pin terminal block. The second pin is not connected.



Use heat shrink tube (black recommended) for the neon (minus sign). I had to extend the wires a little bit to get the correct height.



Don't populate the H8 and H9 (MOTION DETECTOR) header on the driver board.

IN-17 Version

Use heat shrink tube (black recommended) for all the neons.